

Cluster: Antenna and Communication



**TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
PULCHOWK CAMPUS**

**Domain-Adaptive Deep Learning for Massive MIMO OFDM
Channel Estimation Using Measured CSI**

Submitted by

Prabin Chandra Gautam (PUL080BEI028)

Saurab Gyawali (PUL080BEI039)

Smriti Khanal (PUL080BEI042)

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ABSTRACT

Massive multiple-input multiple-output (MIMO) and orthogonal frequency-division multiplexing (OFDM) form the backbone of modern and emerging wireless communication systems. Accurate channel state information (CSI) is essential for reliable beamforming, detection, and resource allocation. However, conventional channel estimation techniques such as least squares (LS) and minimum mean square error (MMSE) suffer performance degradation under high mobility, limited pilot resources, and large-scale antenna configurations.

Recent deep learning-based approaches have shown improved performance in CSI estimation but are typically trained on synthetic datasets, leading to poor generalization when applied to real-world measurements due to domain mismatch. To address this limitation, this project proposes a physics-guided, domain-adaptive framework that integrates wireless channel characteristics with self-supervised learning techniques to improve robustness across diverse environments.

The proposed approach leverages simulated CSI data for initial training and incorporates adaptation mechanisms for unlabeled real-world CSI. The performance of the framework is evaluated using metrics such as normalized mean square error (NMSE) and bit error rate (BER), with a focus on improving generalization across different channel conditions.

Keywords: Channel State Information, Massive MIMO, OFDM, Deep Learning, Domain Adaptation, CSI Estimation.

Contents

Abstract	i
Abbreviations	v
1. INTRODUCTION	1
1.1 Background	1
1.1.1 OFDM, Massive MIMO, and Channel Estimation	1
1.2 Motivation	1
1.3 Project Applications	2
2. PROBLEM STATEMENT	2
3. OBJECTIVES	3
3.1 Main Objective	3
3.2 Specific Objectives	3
4. LITERATURE REVIEW	3
4.1 Review of Related Theory	3
4.1.1 OFDM and Massive MIMO	3
4.1.2 Channel Estimation	4
4.1.3 Deep Learning in Wireless Communications	4
4.1.4 Physics-Informed Machine Learning	4
4.1.5 Maximum Mean Discrepancy (MMD)	5
4.2 Review of Related Works	5
4.3 Gap Analysis	6
5. PROPOSED METHODOLOGY	7
5.1 Research Design	7
5.2 System Architecture	8
5.3 Algorithms and Models to be Used	9
5.3.1 Baseline Deep Learning Models	9
5.3.2 Physics-Guided Loss Function	9
5.3.3 Self-Supervised Domain Adaptation	10
5.4 Dataset and Data Acquisition Plan	10
5.4.1 Synthetic Datasets	10
5.4.2 Real CSI Dataset	11
6. TIME SCHEDULE	11
7. EXPECTED OUTCOMES	11

List of Figures

1	Multi-stage framework for physics-guided domain-adaptive CSI estimation	8
2	System architecture of the proposed CSI estimation framework	9

List of Tables

1	Gap Analysis of Existing Works	6
2	6-Week Project Timeline	11

ABBREVIATIONS

5G	Fifth Generation
6G	Sixth Generation
BER	Bit Error Rate
CDL	Clustered Delay Line
CNN	Convolutional Neural Network
CSI	Channel State Information
FSRCNN	Frequency-Selective Residual CNN
LS	Least Squares
MIMO	Multiple-Input Multiple-Output
MMSE	Minimum Mean Square Error
NMSE	Normalised Mean Square Error
OFDM	Orthogonal Frequency Division Multiplexing

1. INTRODUCTION

1.1 Background

Massive multiple-input multiple-output (Massive MIMO) combined with orthogonal frequency division multiplexing (OFDM) forms the core air-interface technology of fifth-generation (5G) wireless systems and is a key enabler for sixth-generation (6G) networks. The combination of large-scale antenna arrays and multicarrier transmission provides significant gains in spectral efficiency, spatial multiplexing, and link reliability. These gains, however, rely on accurate channel state information (CSI) at the receiver, which is essential for beamforming, multi-user detection, link adaptation, and resource allocation.

1.1.1 OFDM, Massive MIMO, and Channel Estimation

Orthogonal frequency division multiplexing (OFDM) enables reliable transmission over frequency-selective channels by dividing the spectrum into orthogonal subcarriers. A cyclic prefix is used to mitigate inter-symbol interference caused by multipath propagation, provided it exceeds the channel delay spread.

Massive multiple-input multiple-output (Massive MIMO) systems employ a large number of base-station antennas to improve spectral efficiency and spatial diversity. In time-division duplex (TDD) operation, uplink pilots are used to estimate downlink channel state information (CSI) through channel reciprocity, reducing pilot overhead compared to frequency-division duplex (FDD) systems. Accurate CSI is essential for beamforming, multi-user detection, and resource allocation.

Channel estimation refers to recovering the channel frequency response from received pilot signals. Conventional methods include Least Squares (LS), which does not exploit channel statistics, and Minimum Mean Square Error (MMSE) and its linear approximation (LMMSE), which utilize second-order channel statistics for improved accuracy. However, these statistical assumptions are often unreliable in dynamic environments, and performance degrades under limited pilots, mobility, and large antenna arrays.

1.2 Motivation

Accurate channel state information (CSI) is fundamental for the performance of Massive MIMO–OFDM systems. While traditional estimation methods provide reliable performance under ideal assumptions, their effectiveness degrades in practical deployments due to environmental variability and modeling limitations. This motivates the need for data-driven approaches that can generalize across different wireless conditions while

leveraging inherent channel structure.

These challenges motivate the development of a domain-adaptive framework that incorporates wireless channel structure as inductive bias and enables learning from unlabeled real CSI through self-supervision, improving robustness across heterogeneous environments.

1.3 Project Applications

The proposed framework can be applied in the following areas:

- **5G and emerging 6G systems:** Improving CSI estimation accuracy in Massive MIMO–OFDM networks.
- **Beamforming and spatial multiplexing:** Enabling more reliable precoding and multi-user communication.
- **Real-world wireless deployments:** Enhancing robustness across diverse propagation environments and channel conditions.

2. PROBLEM STATEMENT

Current deep learning-based CSI estimation methods suffer from several key limitations:

1. **Heavy dependence on synthetic training datasets:** Most models are trained on simulated channel data, which does not fully capture real-world propagation effects.
2. **Poor generalisation to real measured environments:** The domain gap between synthetic and real CSI leads to significant performance degradation when models are deployed in practical systems.
3. **Lack of physical interpretability:** Many deep learning models operate as black boxes and do not explicitly incorporate known wireless channel properties such as sparsity and spatial correlation.
4. **Requirement of labelled channel measurements:** Supervised learning approaches rely on ground-truth CSI, which is expensive and difficult to obtain in real deployments.

Therefore, there is a need for a CSI estimation framework that can effectively generalize from simulated to real environments while incorporating wireless channel structure and reducing dependence on labelled data.

3. OBJECTIVES

3.1 Main Objective

The main objective of this project is to develop a channel-aware, domain-adaptive deep learning framework for Massive MIMO–OFDM channel state information (CSI) estimation, leveraging inherent wireless channel properties to enable robust generalization from simulated to real measured environments.

3.2 Specific Objectives

The specific objectives of the project are listed below:

1. To understand the characteristics and challenges of channel estimation in Massive MIMO–OFDM systems.
2. To explore deep learning techniques for improving domain generalization in channel estimation across diverse wireless channel environments.
3. To analyze the effectiveness of the developed framework using simulated and real-world wireless channel data.

4. LITERATURE REVIEW

4.1 Review of Related Theory

4.1.1 OFDM and Massive MIMO

Orthogonal Frequency Division Multiplexing (OFDM) divides the available bandwidth into N orthogonal subcarriers, converting a frequency-selective fading channel into a set of flat-fading subchannels. A cyclic prefix (CP) of length greater than the maximum delay spread eliminates inter-symbol interference, yielding the per-subcarrier input–output model:

$$Y_k = H_k X_k + W_k, \quad k = 0, 1, \dots, N - 1 \quad (4.1)$$

where H_k , X_k , and $W_k \sim \mathcal{CN}(0, \sigma_w^2)$ are the channel frequency response, transmitted symbol, and noise on subcarrier k , respectively.

Massive MIMO equips the base station with $M \gg K$ antennas to serve K users simultaneously. The received signal vector $\mathbf{y} \in \mathbb{C}^M$ is:

$$\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{w} \quad (4.2)$$

where $\mathbf{H} \in \mathbb{C}^{M \times K}$ is the MIMO channel matrix. As $M \rightarrow \infty$, channel hardening ensures $\frac{1}{M} \mathbf{H}^H \mathbf{H} \rightarrow \mathbf{I}_K$, enabling simple linear processing and high spectral efficiency.

4.1.2 Channel Estimation

Pilot-aided channel estimation observes $\mathbf{Y}_p = \mathbf{H}_p \mathbf{X}_p + \mathbf{W}_p$ at known pilot subcarriers. The three classical estimators are:

- **Least Squares (LS):** Minimises residual power with no channel statistics required:

$$\hat{\mathbf{H}}_{\text{LS}} = \mathbf{Y}_p / \mathbf{X}_p \quad (4.3)$$

Simple to implement but noise-sensitive, with $\text{MSE} \propto \sigma_w^2$.

- **MMSE:** Incorporates the channel covariance $\mathbf{R}_H = \mathbb{E}[\mathbf{h}\mathbf{h}^H]$ to minimise estimation error:

$$\hat{\mathbf{H}}_{\text{MMSE}} = \mathbf{R}_H (\mathbf{R}_H + \sigma_w^2 \mathbf{I})^{-1} \mathbf{H}_{\text{LS}} \quad (4.4)$$

Optimal in the MSE sense but requires knowledge of channel statistics.

- **LMMSE:** Approximates MMSE using a simplified or estimated $\mathbf{R}_H$, balancing performance and computational cost.

Performance of all three degrades with high mobility and sparse pilot grids, motivating data-driven approaches.

4.1.3 Deep Learning in Wireless Communications

Deep learning approaches formulate channel estimation as a supervised learning problem, where a neural network learns to map received pilot signals to the corresponding channel state information (CSI).

Convolutional neural networks (CNNs) exploit local correlations in the time-frequency domain using learned filters, making them effective for capturing structured patterns in wireless channels. Residual networks (ResNets) improve training stability and performance by enabling deeper architectures through skip connections.

These data-driven methods are capable of learning complex, non-linear channel relationships and generally outperform traditional estimators, particularly in challenging and time-varying wireless environments.

4.1.4 Physics-Informed Machine Learning

Physics-informed machine learning integrates prior knowledge of wireless channel structure into data-driven models to improve generalization and reduce data dependency.

In this work, the following physical priors are considered:

- **Delay-domain sparsity:** Wireless channels typically consist of a small number of dominant multipath components. This sparsity is exploited by encouraging sparse representations of the channel in the delay domain.
- **Spatial correlation:** Signals received across antenna elements in Massive MIMO systems exhibit correlation due to shared scattering environments. This property is used to constrain channel estimates within physically consistent spatial subspaces.

4.1.5 Maximum Mean Discrepancy (MMD)

Maximum Mean Discrepancy (MMD) is a statistical distance metric used to measure the difference between two probability distributions based on their feature embeddings. It is widely used in domain adaptation to reduce distribution mismatch between source and target domains.

Given two sets of feature representations $\{x_i\}_{i=1}^n$ from the source domain and $\{y_j\}_{j=1}^m$ from the target domain, MMD is defined as:

$$\text{MMD}^2 = \left\| \frac{1}{n} \sum_{i=1}^n \phi(x_i) - \frac{1}{m} \sum_{j=1}^m \phi(y_j) \right\|_{\mathcal{H}}^2 \quad (4.5)$$

where $\phi(\cdot)$ represents a mapping to a reproducing kernel Hilbert space (RKHS), and $\mathcal{H}$ denotes the associated Hilbert space.

In wireless communication systems, MMD is useful for aligning feature distributions between synthetic CSI (source domain) and real measured CSI (target domain).

4.2 Review of Related Works

The field of OFDM channel estimation has evolved from classical statistical approaches toward deep learning-based frameworks. Traditional pilot-aided channel estimation methods primarily rely on Least Squares (LS) and Minimum Mean Square Error (MMSE) estimators. While LS is simple to implement, it is highly sensitive to noise. MMSE improves estimation accuracy by exploiting channel statistics but requires prior covariance knowledge and computationally expensive matrix inversions [1, 2, 3, 4].

Recent studies formulate channel estimation as a learning problem by treating the time-frequency channel response as a two-dimensional structured signal. This enables the use of convolutional neural networks (CNNs) to exploit local correlations. For example,

ChannelNet employs cascaded CNNs to refine coarse LS estimates into high-resolution channel reconstructions [1, 5, 6].

For millimeter-wave (mmWave) and massive MIMO systems, deep learning has also been applied to hybrid analog-digital architectures. Elbir et al. proposed deep neural networks for joint channel estimation and hybrid beamforming, directly mapping received pilot signals to beamforming matrices [7]. Other approaches, such as LDAMP and deep learning-based compressed sensing (DLCS), combine sparse signal recovery with neural-network-based denoising and precoding [8].

To model temporal channel variations, recurrent and attention-based architectures have been investigated. GRU and LSTM networks capture channel dynamics without explicit statistical models, while attention mechanisms improve estimation by focusing on relevant spatial and spectral regions [4, 9, 3, 6].

Recent work has further explored alternative signal representations and end-to-end learning. Wavelet-based approaches improve robustness to noise and interference [10], whereas some methods directly learn mappings from received signals to transmitted symbols without explicit channel estimation [2]. To support benchmarking and reproducibility, the DeepMIMO dataset provides a ray-tracing-based framework for generating realistic MIMO channels across diverse propagation environments [11].

4.3 Gap Analysis

Based on the literature review, the following gaps are identified:

Table 1: Gap Analysis of Existing Works

Existing Work	Limitation
CNN/ResNet-based CSI Estimation	Primarily trained and evaluated on synthetic datasets, resulting in limited generalization to real measured channels.
MMSE and Statistical Estimators	Require accurate channel statistics and become computationally expensive in large-scale Massive MIMO systems.
Deep Learning Channel Estimation	Often operates as a black box and does not explicitly exploit channel sparsity and spatial correlation.
Domain Adaptation Methods	Limited work addresses adaptation from simulated CSI to real measured CSI without labeled target-domain data.

Most existing CSI estimation frameworks rely heavily on simulated channel models and

exhibit performance degradation when deployed in real wireless environments. Furthermore, many deep learning approaches lack physical interpretability and require large amounts of labeled data for training. These limitations motivate the development of a physics-guided self-supervised domain adaptation framework capable of transferring knowledge from synthetic datasets to real measured CSI while preserving underlying wireless channel characteristics.

5. PROPOSED METHODOLOGY

This chapter describes the proposed research design, system architecture, algorithms, and dataset plan for the physics-guided domain-adaptive CSI estimation framework.

5.1 Research Design

The proposed framework is organised into five stages, as illustrated in Figure 1.

1. **Stage 1 – Synthetic and Measured CSI Dataset Preparation:** Synthetic Massive MIMO–OFDM channel datasets are generated using established models such as DeepMIMO and 3GPP CDL to represent diverse propagation environments. In addition, real measured CSI data is collected and preprocessed to serve as the target domain for domain adaptation.
2. **Stage 2 – Baseline Channel Estimation:** Conventional estimators, including Least Squares (LS) and Minimum Mean Square Error (MMSE), are implemented as analytical baselines. A lightweight CNN-based channel estimator is also developed and trained on synthetic CSI to learn a data-driven mapping from received pilots to channel estimates.
3. **Stage 3 – Physics-Guided Channel Learning:** The CNN estimator is enhanced using wireless channel structure priors. Specifically, delay-domain sparsity is enforced by applying an L_1 penalty on the sparse representation of the channel, while spatial correlation across antenna elements is preserved using a correlation-based regularization term. These constraints guide the model toward physically consistent channel estimates and improve generalisation.
4. **Stage 4 – Self-Supervised Domain Adaptation:** To reduce dependence on labelled real CSI, domain adaptation is performed using unlabeled measured data. Maximum Mean Discrepancy (MMD) is used to align feature distributions between synthetic and real domains, while consistency regularization enforces stable predictions under input perturbations such as noise injection and pilot masking. This enables effective transfer from simulation to real environments.

5. **Stage 5 – Performance Evaluation and Analysis:** The proposed framework is evaluated using synthetic and real CSI datasets. Performance metrics include Normalised Mean Square Error (NMSE), Bit Error Rate (BER), generalisation gap between domains, and adaptation gain over direct transfer learning. Comparisons are made against LS, MMSE, and the baseline CNN model to quantify improvements from physics guidance and domain adaptation.

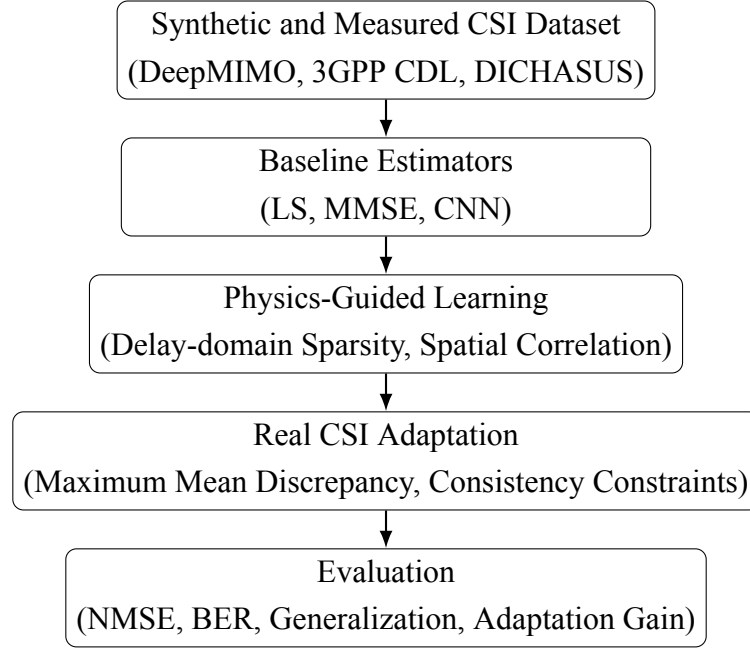


Figure 1: Multi-stage framework for physics-guided domain-adaptive CSI estimation

5.2 System Architecture

The proposed system is designed as a modular processing pipeline for Massive MIMO–OFDM CSI estimation and domain adaptation. The data flow through the system is sequential, as described below.

1. **OFDM Receiver Front-End:** The received time-domain OFDM signal is processed to remove cyclic prefix and perform FFT, converting it into the frequency-domain received signal.
2. **Pilot Processing Module:** Pilot subcarriers are extracted from the received OFDM frame based on the predefined pilot pattern. These pilot observations are used for initial channel estimation.
3. **Initial Channel Estimation Module (LS/MMSE):** A conventional estimator (Least Squares or MMSE) generates a coarse estimate of the channel at pilot and interpolated subcarrier locations. This forms the initial CSI input to the learning system.

4. **CSI Representation Module:** The estimated channel is reshaped into a structured Massive MIMO tensor representation of dimensions (antennas $\times$ subcarriers $\times$ real/imaginary components), which serves as input to the deep learning model.
5. **Deep Learning Inference Module:** A lightweight CNN-based model processes the CSI tensor and outputs a refined full-band channel estimate across all antennas and subcarriers.
6. **Domain Adaptation Module:** During deployment with real measured CSI (DICHASUS dataset), feature representations from the CNN are aligned with the synthetic domain using statistical distribution matching (MMD-based alignment). This module operates on extracted feature embeddings.
7. **Output Reconstruction Module:** The final predicted CSI is reconstructed into complex-valued channel matrices for downstream tasks such as channel equalization and performance evaluation.

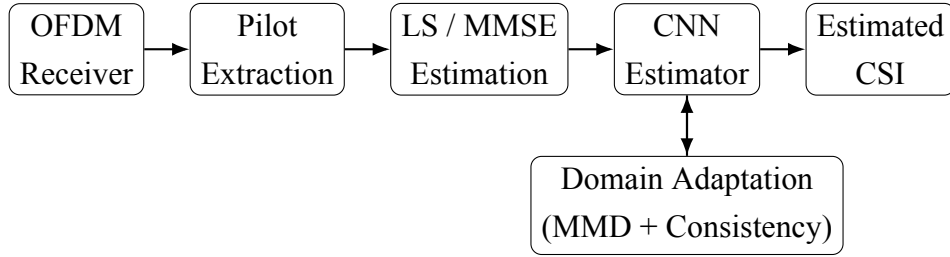


Figure 2: System architecture of the proposed CSI estimation framework

5.3 Algorithms and Models to be Used

5.3.1 Baseline Deep Learning Models

We will implement and compare the following architectures:

- **CNN:** A convolutional neural network for CSI feature extraction.
- **ResNet:** A residual learning-based extension of CNN for deeper feature representation.

5.3.2 Physics-Guided Loss Function

The training objective integrates data-driven reconstruction with wireless channel structure priors. The total loss is defined as:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{MSE}} + \lambda_{\text{sparse}} \mathcal{L}_{\text{sparse}} + \lambda_{\text{corr}} \mathcal{L}_{\text{corr}} \quad (5.1)$$

where:

- $\mathcal{L}_{\text{MSE}}$ measures reconstruction error between estimated and ground-truth CSI on synthetic data.
- $\mathcal{L}_{\text{sparse}}$ enforces sparsity in the delay-domain representation of the channel using an L_1 penalty, reflecting the multipath nature of wireless propagation.
- $\mathcal{L}_{\text{corr}}$ preserves spatial correlation across antenna elements to maintain physically consistent Massive MIMO channel structure.

5.3.3 Self-Supervised Domain Adaptation

For real measured CSI, adaptation is performed without requiring ground-truth labels using the following mechanisms:

- **Maximum Mean Discrepancy (MMD):** Reduces the distribution gap between synthetic and real CSI by aligning feature-level statistics across domains.
- **Consistency Regularisation:** Ensures stable predictions by enforcing similar outputs for perturbed inputs, where perturbations include additive noise and pilot masking applied to the input CSI.

These components are integrated into the overall training objective, resulting in the final loss function:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{MSE}} + \lambda_{\text{sparse}}\mathcal{L}_{\text{sparse}} + \lambda_{\text{corr}}\mathcal{L}_{\text{corr}} + \lambda_{\text{MMD}}\mathcal{L}_{\text{MMD}} + \lambda_{\text{cons}}\mathcal{L}_{\text{consistency}} \quad (5.2)$$

where

- $\mathcal{L}_{\text{MMD}}$: Measures the statistical distance between feature distributions of synthetic and real CSI data.
- $\mathcal{L}_{\text{consistency}}$: Encourages invariant predictions for perturbed versions of the same input CSI sample, i.e., $f(x) \approx f(\tilde{x})$.

5.4 Dataset and Data Acquisition Plan

5.4.1 Synthetic Datasets

We will generate synthetic CSI data using the following models:

- **DeepMIMO:** A widely used ray-tracing dataset for massive MIMO channels.
- **3GPP CDL:** Clustered delay line models for 5G NR evaluation.

5.4.2 Real CSI Dataset

For real-world evaluation and domain adaptation, we use over-the-air measured Massive MIMO CSI from the DICHASUS dataset.

- **DICHASUS CSI Dataset (DICHASUS-005x):** Contains measured Massive MIMO-OFDM channel responses collected using a 32-antenna array with 1024 subcarriers.

6. TIME SCHEDULE

The proposed project is planned over a 6-week period, as detailed in Table 2.

Table 2: 6-Week Project Timeline

Week	Activities
Week 1	Literature review, dataset collection (DeepMIMO, 3GPP CDL, real CSI dataset)
Week 2	Implementation of LS and MMSE estimators; initial data preprocessing
Week 3	Development of CNN baseline model and training on synthetic data
Week 4	Design and integration of physics-guided loss functions
Week 5	Implementation of self-supervised adaptation on real CSI data
Week 6	Comprehensive experiments (NMSE, BER, generalisation), report writing, and presentation preparation

7. EXPECTED OUTCOMES

Upon completion of this project, the following outcomes are anticipated:

1. **Unified CSI estimation framework:** A complete deep learning-based pipeline for channel state information estimation in Massive MIMO–OFDM systems.
2. **Domain-adaptive learning approach:** A model capable of improving generalization across different wireless channel environments, including simulated and real-world data.

3. **Physics-aware deep learning model:** An architecture that exploits structural properties of wireless channels, including sparsity and spatial-frequency correlations, to enhance robustness in CSI estimation.
4. **Improved estimation performance:** Demonstrated enhancement in CSI estimation accuracy compared to conventional baseline methods.
5. **Comprehensive experimental evaluation:** A systematic assessment of the proposed approach under multiple channel conditions using standard performance metrics.

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